

Stoquify POS and Sales-to-Cash Master Implementation Roadmap

Roadmap date: 2026-08-17

Program status: BLOCKED — implementation and production evidence required

Planning horizon: 40–44 weeks with two dedicated squads; 52–64 weeks with one squad

Scope: immediate POS, electronic tenders, receipts/fiscal evidence, store close, delivery/on-account order-to-cash, controls, operations, and release proof

Classification: implementation program; not a legal, accounting, PCI, security, accessibility, fiscal, or production certification

1. Purpose and roadmap authority

This document consolidates the current POS audit, enterprise Sales-to-Cash audit, readiness assessment, implementation roadmap, evidence record, and the historical AqStoqFlow implementation-gap sequence into one implementation program.

It does not replace the underlying reports. It provides the execution order, work packages, ownership, dependencies, migrations, gates, acceptance evidence, and stop conditions needed to implement their proposals without architectural guesswork.

Primary sources:

- STOQUIFY_POS_ENTERPRISE_GRADE_AUDIT_2026-08-16.md
- STOQUIFY_POS_ENTERPRISE_GRADE_AUDIT_EVIDENCE_2026-08-16.md
- STOQUIFY_ENTERPRISE_SALES_TO_CASH_AUDIT_AND_MODERNIZATION_2026-08-16.md
- STOQUIFY_ENTERPRISE_SALES_TO_CASH_READINESS_2026-08-16.md
- STOQUIFY_ENTERPRISE_SALES_TO_CASH_IMPLEMENTATION_ROADMAP_2026-08-17.md
- Historical AQSTOQFLOW_JUNE_23_24_IMPLEMENTATION_GAP_PLAN_2026-06-24.md and its skills-suite proposal, used only for sequencing and evidence discipline
- graphify-out component, action, app, and hook topology used by the source audits

Where sources disagree, the newer repository-backed Sales-to-Cash reports govern current product truth. Completed controls stay closed unless a regression is reproduced. The historical gap plan governs verification-first sequencing and enforcement discipline, not stale implementation status.

2. Program decision

Stoquify will deliver two coordinated product paths:

1. Harden the existing immediate POS path until sale result, cash/electronic tender, stock, accounting, receipt/fiscal source, shift close, offline recovery, devices, and operator experience remain trustworthy under concurrency and failure.
2. Build a separate delivery/on-account order-to-cash orchestrator for promise, reservation, fulfillment, physical goods issue, invoicing, AR, collections, returns, and corrections.

Both paths reuse the existing inventory stock-event/valuation, accounting posting, payment/provider, reconciliation, receipt/fiscal, audit, close-invalidation, and outbox kernels.

The following are architectural invariants:

- services/pos/pos.service.ts::commitPOSSale remains the only immediate-sale finalizer.
- Offline replay enters the same finalizer and never creates a parallel sale kernel.
- Delivery order confirmation is non-posting.
- Reservation changes available quantity, not on-hand quantity or COGS.
- Physical goods issue moves inventory and COGS exactly once.
- Invoice eligibility and AR follow the approved billing event, not an arbitrary status label.

- Provider pending, unknown, timeout, or operator-entered reference never becomes successful payment.
- A completed sale binds immutable receipt/fiscal-source evidence before returning its result.
- Receipt delivery failure never reverses or conceals financial completion.
- Original financial, stock, receipt, fiscal, and audit facts are corrected with linked compensating records, never mutation.
- Client previews never own price, tax, discount, stock, tender, refund, or cash truth.

3. Definition of ambition fulfilled

The system ambition is fulfilled only when all of the following are true for an explicitly supported country, currency, provider, hardware, tax, inventory, and receipt/fiscal configuration:

1. Every material transition has one authoritative owner and versioned state contract.
2. Identical retries return one immutable original result; conflicting retries fail closed and audit.
3. Every electronic payment is provider-authoritative from initiation through settlement, reversal, dispute, and chargeback.
4. Every completed sale has immutable receipt/fiscal-source evidence independent of delivery availability.
5. Immediate POS and delivery/on-account workflows reuse kernels without sharing one overloaded lifecycle.
6. Stock quantity, reservation, valuation, COGS, revenue, tax, clearing, AR, fees, refunds, and corrections occur at approved events and tie out.
7. Shift close, tender declaration, business-day statement, provider reconciliation, and accounting close are distinct, observable, and governed.
8. Navigation, route data, reads, writes, offline commands, receipts, reports, exports, and APIs enforce consistent tenant, location, RBAC, and entitlement policy.
9. Cashier, manager, accountant, inventory, treasury, support, and approver workflows are recoverable, bilingual, responsive, accessible, and privacy-minimized.
10. Offline, hardware, browser, provider, fiscal worker, concurrency, load, restore, and failure behavior are proven in the named release environment.
11. Every gate is bound to one commit, migrations, configuration, environment, evidence index, and reviewer decision.
12. Unsupported combinations fail closed and are not marketed as supported capability.

4. Delivery model

4.1 Teams

Team	Primary ownership
Squad A — Transaction Trust and Payments	Result registry, cart concurrency, payment/provider bridge, receipts/fiscal source, reconciliation, immediate-sale backend
Squad B — Store Operations and Order-to-Cash	POS workstation, shift/business day, reservation, fulfillment, invoice/AR, returns, operator workflows
Shared Platform	RBAC/entitlement, module control, country packs, design system, telemetry, CI/release, migration tooling
Control Review Board	Product, accounting, inventory, treasury/payments, security, country-pack, accessibility, operations/support, SRE/release

4.2 Capacity assumptions

- Two squads of 5–7 engineers each, including a senior backend owner and senior frontend owner.
- Dedicated product manager and program manager.
- Named accounting and inventory controllers with scheduled weekly decision capacity.
- Security, accessibility, SRE, country-pack, and payments/provider reviewers are available at defined gates.
- Real PostgreSQL integration environment is ready in Mobilization.

- Provider sandbox, supported hardware, and selected pilot country configuration are secured before their dependent phases.

Adding developers without decision capacity, realistic environments, and controller review will not shorten the critical path.

5. Program governance

5.1 Required registers

Maintain these versioned artifacts from week 1:

- Decision register and architecture decision records.
- Requirement and finding traceability matrix.
- State-transition and command catalog.
- Migration/backfill register.
- Permission, entitlement, fresh-auth, and maker-checker matrix.
- Country/provider/hardware capability matrix.
- Risk and dependency registers.
- Evidence index tied to commit, environment, configuration, and gate.
- Runbook and operational ownership catalog.
- Release cohort and rollback register.

5.2 Change and evidence rules

- Use bounded implementation slices with declared file ownership.
- Capture git status before every slice and preserve unrelated dirty-worktree changes.
- Use additive expand/backfill/verify/enforce/contract migrations; never reset business data.
- Every slice leaves a dated run report under what-next with files changed, tests, failures, residual risk, and next handoff.
- Static/source evidence, unit/mock evidence, database/runtime evidence, provider/hardware/browser evidence, staging evidence, pilot evidence, and qualified-human review remain visibly distinct.
- A gate outcome is PASS, PASS WITH EXPLICIT LIMITED SCOPE, or BLOCKED. "Code complete" is not a gate outcome.
- No enforcement or external-readiness claim advances without rollback and authenticated runtime evidence.

5.3 Remedy classification

Classification	Program use
Surgical repair	XAF precision, seed least privilege, unsupported tender visibility, localized copy, privacy masking
Additive refactor	Result registry, cart version/idempotency, permission normalization, immutable source, correlation contract
Bridge/adaptor	POS-to-provider payment, device adapters, country-pack fiscal adapter, offline command integration
New aggregate/rebuild	Business day/statement, delivery order, reservation, fulfillment, invoice/AR orchestration
Controlled UI rebuild	POS workstation structure behind frozen server contracts
Hardening/refactor	Component/service decomposition, SLOs, runbooks, partial-data envelopes, load/restore evidence
Blocked	Provider live proof, country-pack conclusions, physical device proof, certifications without required external evidence
Preserve/no-op	Existing one-transaction sale finalizer, accounting posting kernel, inventory stock-event kernel, full refund/void compensation, offline replay kernel, audit/outbox, signed receipt-token foundation unless regression appears

6. Dependency roadmap

flowchart LR

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M0["M0 Verification foundation"] --> M1["M1 Contract and control freeze"]
M1 --> M2["M2 Transaction, cart and access trust"]
M2 --> M3A["M3A Provider-authoritative payments"]
M2 --> M3B["M3B Immutable receipt/fiscal source"]
M3A --> M4["M4 Enterprise POS workstation"]
M3B --> M4
M4 --> M5["M5 Business day, statements and close"]
M2 --> M6["M6 Delivery order and fulfillment"]
M6 --> M7["M7 Invoice, AR, returns and corrections"]
M5 --> M8["M8 Unified operations and assurance"]
M7 --> M8
M8 --> M9["M9 Country pilot and expansion"]

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Critical path: M0 → M1 → M2 → M3A/M3B → M4 → M5 → M8 → M9.

M6 may begin after M2 and run in parallel with M3–M5. M7 follows M6. Parallel work is allowed only across frozen interfaces and disjoint ownership. Teams must not independently change sale finalization, payment-state mapping, stock-event semantics, posting rules, or fiscal-source ownership.

7. Milestone plan

M0 — Verification foundation and program mobilization

Indicative duration: weeks 1–2

Primary classification: surgical repair and test infrastructure

Gate: G0 Verifiable Baseline

Objectives

- Make every later slice testable against real database/runtime behavior.
- Establish a clean, attributable implementation baseline without discarding user work.
- Name decision owners, environments, pilot scope, and evidence standards.

Work packages

ID	Deliverable	Owner
M0-01	Program charter, named owners, scope/non-goals, risk/dependency register	Program lead
M0-02	Clean named implementation baseline or isolated worktree; dirty-worktree attribution	Release lead
M0-03	Reusable PostgreSQL integration and concurrency harness	DBA + backend lead
M0-04	Authenticated EN/FR browser harness for POS, reconciliation, close, owner, manager, and Cash Command	QA lead
M0-05	Responsive viewport, keyboard, accessibility, and non-sensitive screenshot harness	Accessibility + QA
M0-06	Migration/runtime-table verification and fail-fast release check	DBA + SRE
M0-07	Evidence taxonomy, artifact schema, redaction scan, and immutable result storage for gates	SRE + security
M0-08	Baseline typecheck, build, focused tests, policy gates, graph snapshot, and current failures	Release lead
M0-09	Select pilot country, XAF currency profile, provider sandbox, locations, terminal and hardware set	Product + operations

G0 exit criteria

- Full build and baseline gates complete or every failure has an owner and blocker record.
- Authenticated browser smoke reaches the named tenant routes; unauthenticated redirects are not accepted as smoke proof.
- PostgreSQL concurrency tests can run without destructive database operations.
- Required runtime tables/migrations are verified by normal tooling or a documented guarded checker.
- Evidence artifacts identify commit, environment, configuration, timestamp, command, output, and redaction status.
- Pilot country/provider/hardware scope and required reviewers are named.

Stop conditions

Do not begin schema enforcement, provider integration, module hard enforcement, or external-readiness claims if the migration, database, browser, and build foundations are not reproducible.

M1 — Contract, capability, and control freeze

Indicative duration: weeks 3–4

Primary classification: design and policy freeze

Gate: G1 Architecture and Control Freeze

Work packages

ID	Deliverable	Owner
M1-01	Versioned state catalog for sale, cart, provider payment, receipt/fiscal source, session, business day, statement, order, reservation, fulfillment, invoice, allocation, return, and correction	Architecture
M1-02	Command template covering actor, permission, entitlement, precondition, transaction, idempotency, event, stock/accounting consequence, audit, recovery, and terminal result	Architecture + controls
M1-03	Canonical XAF/minor-unit, rounding, tax snapshot, quantity precision, date/time, and business-day contract	Money + country-pack
M1-04	Tender capability matrix with method-specific fields, provider requirement, overpay policy, reconciliation and availability reason	Payments + product
M1-05	Sellable availability, reservation, negative-stock, UOM, lot/serial/expiry and valuation policy	Inventory controller
M1-06	Permission/entitlement/fresh-auth/maker-checker matrix for every POS and Sales-to-Cash command/surface	Security + product
M1-07	Receipt/fiscal-source, numbering, correction, offline/provisional and retention contract	Country-pack + accounting
M1-08	Device readiness and offline-support capability matrix; unknown is the default	Operations + platform
M1-09	SLO, privacy-minimization, log redaction, evidence retention and support ownership contract	SRE + security

G1 exit criteria

- Every material transition has one owner and permitted predecessor/successor states.
- All money and quantity fields have authoritative precision and rounding rules.
- Supported tenders, currencies, tax modes, inventory modes, devices, receipt channels and offline commands are explicit.
- Unsupported combinations fail closed.
- Accounting, inventory, payments, security, product, country-pack and operations approve their contracts.
- No national legal rule is hard-coded into shared domain code.

M2 — Transaction, cart, inventory, and access trust

Indicative duration: weeks 5–8

Primary classification: additive refactor and surgical security repair

Gate: G2 Transaction and Access Trust

Workstream A — sale result and cart concurrency

ID	Implementation slice	Owner
M2-A01	Add tenant/terminal-scoped clientCommitId result-registry schema with canonical request hash and schema version	POS backend
M2-A02	Claim → commit/result lifecycle with immutable redacted result envelope	POS backend
M2-A03	Integrate result claim and result persistence into the existing commitPOSSale transaction	POS backend
M2-A04	Return original result for identical retries; reject and audit key/payload conflicts	POS backend
M2-A05	Add one-active-draft identity invariant and unique cart-line identity after measuring existing data	POS backend + DBA
M2-A06	Add cart version and commandId to create/add/update/remove commands	POS backend + frontend
M2-A07	Return authoritative cart and typed stale/conflict result after every mutation	POS backend
M2-A08	Replace silent quantity clamping with sellable-availability conflict and refreshed version	Inventory + POS
M2-A09	Add real PostgreSQL races for duplicate commit, duplicate scan, concurrent cart edits, stock reservation/issue, response loss, rollback, and replay	QA + DBA

Workstream B — access, roles, privacy, and direct boundaries

ID	Implementation slice	Owner
M2-B01	Define one POSSurfaceAccess descriptor consumed by sidebar, route, page data, reads, writes, offline, receipt, report, export, and API boundaries	Security platform
M2-B02	Reconcile sales versus pos module vocabulary and migrate entitlement data before enforcement	Module platform
M2-B03	Replace substring-based cashier seed permissions with explicit least-privilege allowlist and negative assertions	Identity + seed data
M2-B04	Add organization/location/terminal assignment and cross-tenant/cross-location negative matrix	Security + QA
M2-B05	Enforce fresh auth and maker-checker for refund, void, override, payout, material variance, price/discount exception, manual match, reversal, and credit note	Security + domain owners
M2-B06	Create cashier-minimized customer and receipt DTOs; mask contact data, remove revenue, and replace raw URLs with actions	Privacy + POS
M2-B07	Verify signed/expiring/revocable receipt-token foundation and configure release-secret enforcement; do not rebuild it unless regression is found	Security + release
M2-B08	Harmonize any membership-only item/API read paths with explicit read permission	API + security
M2-B09	Add redaction, evidence-retention, abuse/repeated-submission, and safe-error tests	Security

Additive migration sequence

1. Measure duplicate/null draft and line candidates without changing data.
2. Add nullable result/cart-version/command metadata and new registry tables.
3. Deploy tolerant readers and shadow result/cart writes.
4. Backfill only evidence-supported references; mark unverifiable legacy facts explicitly.
5. Verify counts, hashes, uniqueness candidates, tenant scope, and rollback behavior.
6. Enable new writers per internal cohort.
7. Add uniqueness/not-null enforcement only after verification.
8. Retain compatibility reads until the final pilot gate; never delete immutable evidence during rollback.

G2 exit criteria

- Concurrent identical commit requests create one sale, payment/AR set, stock consequence, drawer/session consequence, journal set, receipt/fiscal source reference, and immutable result.
- A repeated identical request returns the original result; a conflicting payload rejects and audits.
- Concurrent cart mutations either serialize by version or return a typed conflict; no silent lost update.
- Catalog, add, update, preflight, and commit use one sellable-availability contract while final stock CAS remains the last defense.
- Route, navigation, reads, commands, offline, receipts, reports, exports and APIs reach the same permission/entitlement decision.
- Seeded cashier has no accounting posting, supplier payment, administration, payroll, or unapproved sensitive permission.
- Negative tenant/location/terminal and stale-auth tests pass.
- Public/customer data is minimized and no secret or unnecessary PII appears in logs/evidence.

M3A — Provider-authoritative electronic payments

Indicative duration: weeks 9–14

Primary classification: bridge/adapter

Gate: G3A Payment Truth

Work packages

ID	Implementation slice	Owner
M3A-01	Versioned provider-to-canonical state mapping including initiated, pending, authorized, captured, unknown, declined, cancelled, expired, settled, reversed, refunded, disputed and chargeback	Payments
M3A-02	Payment-intent command with internal and provider idempotency	Payments
M3A-03	Signed webhook and polling ingestion with timestamp, replay, payload-hash, redaction and monotonic-state controls	Payments + security
M3A-04	POS tender capability/read model returning enabled methods, disabled reason, required fields and provider choices	Payments + POS
M3A-05	Mobile Money provider and transaction-reference UI/service contract	Payments + POS UI
M3A-06	Hide or explicitly disable Store Credit until liability ledger, balance ownership and correction rules pass their gate	Product + accounting
M3A-07	Checkout bridge that completes sale only after provider-authoritative capture	Payments + POS
M3A-08	Captured-without-sale, late-capture, duplicate, mismatch and unknown-payment case workflow	Treasury + support

ID	Implementation slice	Owner
M3A-09	Provider clearing, settlement, fee, reversal, refund, dispute, chargeback and suspense postings	Accounting + payments
M3A-10	Link provider event, payment transaction, POS sale, ledger source, statement line, match and reconciliation certificate	Payments + reconciliation
M3A-11	Add payment.transaction proof launch in the reconciliation workbench using existing proof contracts	Reconciliation UI
M3A-12	ProviderAccountHealth read model for credential/setup state, callback lag, statement freshness, unresolved exceptions and settlement account	Payments + SRE
M3A-13	Run dedupe/concurrency guard and outage/replay/tamper/stale/high-volume runbooks	Payments + operations

G3A exit criteria

- Pending, unknown, timeout and manual reference never complete a sale.
- Amount, currency, provider account, signature, timestamp and impossible state-transition mismatches quarantine safely.
- Duplicate webhook, poll, callback, user retry and response loss do not duplicate payment, sale, posting or receipt consequences.
- Captured-without-sale, sale-without-settlement, settlement shortfall, fee, reversal, dispute and chargeback remain visible until an owned terminal state.
- Mobile Money captures include provider identity and provider transaction reference.
- Unsupported Store Credit cannot enter commit.
- Provider sandbox evidence covers success, decline, timeout, unknown, duplicate, late capture, reversal, refund, dispute, chargeback, settlement and statement mismatch.
- Reconciliation workbench proof respects read permission, redaction, unavailable, loading and error states.

M3B — Immutable receipt and fiscal-source truth

Indicative duration: weeks 9–13, parallel with M3A

Primary classification: additive refactor and country-pack adapter

Gate: G3B Document Truth

Work packages

ID	Implementation slice	Owner
M3B-01	Versioned immutable receipt/fiscal-source payload covering seller, location, terminal, cashier, lines, prices, taxes, tenders, customer consent and source IDs	Compliance engineering
M3B-02	Materialize and hash source inside the existing sale transaction before result completion	POS + compliance
M3B-03	Resolve the unused synchronous fiscal helper and retain one fiscalization owner	Architecture
M3B-04	Country-pack-governed legal-number allocation and authority submission	Country-pack
M3B-05	Fiscal worker lease, idempotency, retry, backoff, dead-letter, replay and operator recovery	Compliance + SRE
M3B-06	Durable delivery states for each supported print, email, SMS, WhatsApp and public-token channel	Communications + POS
M3B-07	Production token signing secret, scope, expiry, revocation, rotation and release gate	Security + release
M3B-08	Linked void, credit and correction documents that never rewrite the original	Accounting + compliance

ID	Implementation slice	Owner
M3B-09	POS receipt-recovery UI using actions rather than raw URLs	POS UI

G3B exit criteria

- Every completed sale has immutable source payload/hash before its result returns.
- Later catalog, price, tax, customer, organization or location changes cannot alter historical source truth.
- Legal numbering and authority submission fail closed without a qualified country-pack and remain unavailable for unsupported offline allocation.
- Worker retry, dead-letter and replay produce one document/number per authorized scope.
- Delivery failure is visible and recoverable but never changes sale/payment/stock/accounting completion.
- Release-secret enforcement is green; no token, destination, payment secret or unnecessary PII appears in evidence.
- Country-pack decisions are dated, versioned and approved by a qualified human.

M4 — Enterprise POS workstation and cashier operations

Indicative duration: weeks 15–20

Primary classification: controlled UI rebuild and workflow completion behind frozen contracts

Gate: G4 Immediate POS Release Candidate

Workstream A — authoritative surface state

ID	Implementation slice	Owner
M4-A01	POSSurfaceContext read model combining access, location/terminal assignment, shift custody, currency scale, tender capability, device capability and allowed commands	POS backend
M4-A02	Exact barcode lookup separated from debounced text search; both return sellable availability and inventory version	POS + inventory
M4-A03	Preflight command for server totals, stock conflicts, credit rules, receipt readiness and approval requirements	POS backend
M4-A04	Typed command/result envelopes with correlation ID, command ID, aggregate version, financial state, inventory state, payment state, receipt state and safeRetry	POS backend
M4-A05	Replace global POS cache invalidations with targeted authoritative aggregate updates	POS frontend

Workstream B — cashier and exception workflow

ID	Implementation slice	Owner
M4-B01	Rebuild layout around persistent custody/readiness bar, dominant sell surface, continuously visible cart/total/blocker and secondary diagnostics	Product design + UI
M4-B02	Desktop persistent cart rail, tablet split/collapsible pane and mobile sticky Review/Charge plus full-height cart/tender sheet	POS UI
M4-B03	Method-aware tender panel that shows required metadata and never implies payment success before server/provider confirmation	POS UI + payments
M4-B04	Implement or honestly disable park/resume, cancellation, manager override, refund, void and receipt recovery	Product + POS
M4-B05	Inactivity lock, quick re-auth, suspend/handover, pending-cart policy and authorized recovery	Identity + POS
M4-B06	Customer/receipt privacy masking and role-aware secondary detail	Privacy + POS

ID	Implementation slice	Owner
M4-B07	Clear result states: not committed, committed, provider unknown, receipt retry required, stock conflict, stale cart, permission denied and service unavailable	UX + POS

Workstream C — offline and devices

ID	Implementation slice	Owner
M4-C01	Until certified, replace offline-ready language with truthful online-required/degraded states	POS UI
M4-C02	If offline selling is in pilot scope, connect supported commands to signed, ordered, idempotent queue/replay and conflict workbench	Offline platform
M4-C03	Define encrypted local retention, session expiry, device revocation, terminal sequence and provisional receipt policy	Security + offline
M4-C04	Device adapter contract with unknown, connected, ready, degraded, disconnected, stale and last-seen states	Device platform
M4-C05	Supported scanner, printer and drawer adapters with test, retry, fallback and operator guidance	Device platform + operations

Workstream D — accessibility, localization, performance and telemetry

ID	Implementation slice	Owner
M4-D01	Programmatic names, pressed/selected semantics, 44×44 targets, visible focus and deterministic dialog focus restore	Accessibility + UI
M4-D02	Live announcements for scan, cart delta, blocker, payment, receipt, offline and device state without excessive chatter	Accessibility + UI
M4-D03	200% zoom, 320px reflow, reduced motion, non-color status and bright-environment contrast	Design system
M4-D04	Move every POS/offline/device string into EN/FR catalogs and obtain professional French review	Localization
M4-D05	Currency/date/time/quantity formatting from organization and country-pack contracts; XAF inputs accept no fractional francs	Money + UI
M4-D06	Cursor pagination/virtualization, lazy image fallback, precise caching, visibility-aware polling and measured performance budgets	Frontend + API
M4-D07	End-to-end redacted telemetry for scan/search/cart/preflight/approval/tender/commit/receipt/offline/device with safe-retry semantics	SRE + POS

G4 exit criteria

- Cash and every supported electronic tender pass EN/FR browser, database, device, concurrency, failure and recovery scenarios.
- Unsupported tender, currency, tax, stock, offline and hardware capability is hidden or clearly unavailable with a reason.
- XAF values are integer-only from input through storage, posting, receipt and variance.
- Cart total, blocker and next action remain reachable at 1440×1000, 1024×768, 768×1024, 390×844 and 320px/200% zoom.
- Keyboard, screen-reader, touch, focus, contrast and reduced-motion evidence passes the agreed WCAG 2.2 AA target matrix; this remains evidence, not self-certification.
- Lock/handover preserves cashier and cash responsibility; queued data and customer detail are protected.
- Device health comes from adapters with freshness; unknown is never rendered as ready.
- Offline duplicate/conflict cannot silently finalize twice or allocate unsupported final fiscal numbers.
- No secrets, PAN/CVV/PIN, raw provider payloads, raw receipt URLs or unnecessary PII appear in UI, logs, screenshots or support evidence.

M5 — Governed business day, statements, reconciliation, and close

Indicative duration: weeks 21–26

Primary classification: new aggregate and bridge to existing close/reconciliation kernels

Gate: G5 Store Financial Close

Work packages

ID	Implementation slice	Owner
M5-01	Location/time-zone-aware BusinessDay aggregate supporting cross-midnight trading	Store operations + backend
M5-02	Included-session manifest and immutable source hash	POS close + audit
M5-03	Blind cash/tender declaration with denomination and evidence contract	Store operations
M5-04	Variance thresholds, reason codes, fresh-auth approval and no-self-approval	Security + controller
M5-05	Interim X report and immutable final Z report under country-pack terminology	POS close + country-pack
M5-06	Retail statement calculation from included completed transaction/result records	Accounting + POS close
M5-07	Statement posting with source-linked journal evidence and status	Accounting
M5-08	Provider settlement/reconciliation and suspense status in close readiness	Treasury + reconciliation
M5-09	Aggregate fiscal, offline, stock, receipt, posting, entitlement and assurance blockers	Close assurance
M5-10	Accounting-close handoff and invalidation after late posting/correction	Close assurance
M5-11	Complete remaining payroll, compliance, country-pack and entitlement close-invalidation rings	Close assurance + domain owners
M5-12	Compact stale-source drawer and close-readiness journey grouped by typed source	Accounting UI

G5 exit criteria

- Shift/session close, declaration, business day, retail statement, provider reconciliation and accounting close have distinct states, permissions and owners.
- Final statement/Z cannot omit accepted offline events or unresolved conflicts.
- Declared, system, provider, ledger and bank values tie or remain in owned suspense with reason and age.
- Manager approval is attributable, fresh-authenticated and separation-of-duties compliant.
- Late corrections or authority changes invalidate affected evidence and show the typed source.
- X/Z naming, numbering, tax and retention semantics have qualified country-pack review.

M6 — Delivery/on-account order, reservation, and fulfillment

Indicative duration: weeks 21–28, parallel with M5

Primary classification: new orchestration aggregates reusing existing kernels

Gate: G6 Physical Fulfillment Truth

Work packages

ID	Implementation slice	Owner
M6-01	Versioned SalesOrder aggregate with terms, delivery location, price/tax snapshot, credit policy and command idempotency	O2C backend
M6-02	Confirm/cancel commands; confirmation remains non-posting	O2C backend
M6-03	Reservation aggregate with active, partial, consumed, released and expired states	Inventory

ID	Implementation slice	Owner
M6-04	Authoritative sellable availability from on-hand, reserved, policy and location	Inventory
M6-05	Fulfillment aggregate with partial release, pick, pack, ship/handover and delivery evidence	Fulfillment
M6-06	Physical goods issue through the existing inventory stock-event/valuation kernel	Inventory + fulfillment
M6-07	Consume reservation exactly once and preserve backorder quantities	Inventory
M6-08	Cancellation before issue and linked goods-issue reversal after issue	Fulfillment + accounting
M6-09	Operator workbench for exception, partial, damaged, delayed and recovery states	Operations UX
M6-10	Fail closed for unimplemented UOM, lot, serial, expiry, negative-stock or multi-location combinations	Product + inventory

G6 exit criteria

- Order confirmation posts no revenue, AR, COGS or inventory.
- Reservation changes reserved/available but not on-hand or COGS.
- Physical goods issue changes quantity and posts COGS/inventory exactly once.
- Partial fulfillment, backorder, cancellation and reversal preserve original line history.
- Every stock event links order line, fulfillment line, location, valuation evidence, actor and command key.
- Concurrent reservation and issue tests prove no oversell beyond approved policy.
- Unsupported inventory combinations fail closed and are visible to the operator before confirmation.

M7 — Invoice, AR, collections, returns, and corrections

Indicative duration: weeks 29–34

Primary classification: new orchestration reusing accounting/payment/fiscal kernels

Gate: G7 Financial Order-to-Cash

Work packages

ID	Implementation slice	Owner
M7-01	Invoice eligibility from delivered quantity or explicit approved advance policy	Accounting + product
M7-02	Immutable invoice source/fiscal evidence and AR/revenue/tax posting	Accounting
M7-03	Partial invoices, due dates, payment allocations, partial payment and unapplied cash	AR team
M7-04	Provider collections through the M3A payment/reconciliation kernel	Payments + AR
M7-05	Return authorization with received, inspected and disposition states	Returns + inventory
M7-06	Restock, quarantine, damaged and write-off inventory consequences	Inventory controller
M7-07	Source-linked credit note, refund, AR adjustment and goods-issue reversal	Accounting + returns
M7-08	Dispute, chargeback and recovery case without rewriting original transaction	Treasury + support
M7-09	Customer statement, aging, collection evidence and close invalidation	AR + close assurance
M7-10	Role-separated approval UI and evidence timeline for corrections	Product + security

G7 exit criteria

- No quantity is billed twice and only eligible quantity is invoiced.

- Invoice posting balances AR/revenue/tax and binds immutable source/fiscal evidence.
- Customer funds allocate explicitly; unapplied or mismatched cash remains visible.
- Return disposition is approved before sellable stock increases.
- Credit note, refund, stock correction, provider correction and journal reversal link to the original fact.
- Partial fulfillment, invoice, payment and return scenarios tie order, stock, AR, provider and ledger quantities/amounts.
- Maker-checker and fresh-auth policies pass positive and negative tests.

M8 — Unified operations, assurance, maintainability, and hardening

Indicative duration: weeks 35–38

Primary classification: hardening, controlled refactor and observe-mode assurance

Gate: G8 Operational Readiness Candidate

Work packages

ID	Implementation slice	Owner
M8-01	SLOs/error budgets for checkout, provider callback, fiscal/receipt workers, offline replay, close, reconciliation, fulfillment and invoice	SRE
M8-02	Redacted dashboards, alert routes, dead-letter/suspense aging and named ownership	SRE + support
M8-03	Runbook for every queued, unknown, conflict, partial-success and dead-letter state	Operations
M8-04	Load, soak, concurrency, network-loss, provider-outage, worker-backlog and recovery tests	Performance + QA
M8-05	Backup/restore, point-in-time recovery, failover and evidence-restoration drill	DBA + SRE
M8-06	Decompose ProfessionalPOSSystem and pos.service behind frozen state/command contracts; prohibit a second finalizer	Architecture + POS
M8-07	Browser/device/accessibility/visual/performance/localization regression suite	QA + accessibility
M8-08	Security scan, threat-model review, abuse tests, secret rotation, log/evidence and retention review	Security
M8-09	Support timeline linking result, sale, receipt, payment, stock, journal, session, statement, fulfillment, invoice and correction	Support platform
M8-10	Workflow Assurance live observe pilot with seeded incidents, action routes, proof/source hashes and missing-table gate	Assurance platform
M8-11	Finance/command-surface partial-data envelopes and minimized read models	Platform + frontend
M8-12	Stock-to-cash, close-readiness and role-specific daily-habit surfaces grounded in real evidence	Product + analytics
M8-13	Provider health cards and reconciliation runbooks integrated into operations	Payments + SRE

G8 exit criteria

- Every alert has owner, severity, response time, runbook and terminal resolution.
- Restore/failure drills preserve immutable source-linked evidence.
- Performance budgets hold under reference peak and degraded-provider load.
- The decomposed UI/service retains one finalizer and one stock/accounting/fiscal owner per fact.
- One slow non-critical source cannot blank a command surface.
- Observe mode produces real tenant incidents and missing runtime tables fail a release gate rather than a dashboard request.

- Accessibility, localization, supported browser/device, security and privacy regressions are green for the named scope.
- No unresolved critical financial, inventory, payment, isolation, security, accessibility or recovery defect remains.

M9 — Country pilot, narrow enforcement, and production expansion

Indicative duration: weeks 39–44

Primary classification: controlled rollout and evidence-based enforcement

Gate: G9 Production Expansion Decision

Rollout stages

Stage	Cohort	Required entry evidence	Automatic stop/rollback condition
R0	Local and CI	Focused suites, database races, global gates	Any invariant failure
R1	Staging with provider/fiscal sandbox	Full E2E, recovery, accessibility and load matrix	Duplicate consequence or unexplained posting/stock mismatch
R2	Internal staff/demo organization	Supported hardware and controlled business day	False payment success, cross-tenant access or unrecoverable close
R3	One design partner; one country/provider/location set	Qualified approvals and daily evidence review	Any critical financial, fiscal, stock, security, privacy or accessibility issue
R4	Up to 5% eligible cohort	Stable SLOs and no unresolved material suspense	Error-budget, close or support threshold breach
R5	Up to 25% eligible cohort	At least two stable business-day and settlement cycles	Reconciliation drift, evidence loss or support overload
R6	General availability for the explicitly eligible package	Formal G9 decision, support capacity and rollback readiness	Ongoing control or error-budget breach

Enforcement progression

1. Continue module/workflow decisions in observe mode during R0–R2.
2. Measure would-block decisions and eliminate false positives.
3. Select one workflow-assurance check and one POS module surface for a tenant-ring enforcement pilot.
4. Prove seeded failure, owner, corrective route, evidence source/hash, authenticated browser behavior and rollback.
5. Enable only for R3 pilot tenants under feature/ring control.
6. Expand only when denial, correction, support and rollback metrics are stable.
7. Never globally enable broad module enforcement from static inventory alone.

Production evidence pack

- Named commit, build, environment, configuration, migration hashes and deployed country/provider versions.
- Additive migration status, backfill/reconciliation certificates and rollback/disable plan.
- PostgreSQL concurrency, response-loss and failure-injection evidence.
- Provider capture, settlement, fee, reversal, refund, dispute, chargeback and statement proof.
- Receipt/fiscal source, authority, numbering, delivery, secret, retry, dead-letter and correction proof.
- Inventory quantity/valuation/COGS and accounting tie-outs.
- Session, declaration, business day, X/Z, statement, reconciliation and accounting-close proof.
- Tenant/location isolation, RBAC, entitlement, fresh auth, maker-checker, redaction, audit and abuse evidence.
- Browser, accessibility, hardware, offline, performance, restore, failover, monitoring and incident evidence.
- Dated reviewer decisions and explicit limitations.

G9 exit decision

G9 may authorize only the tested country, currency, provider, device, tax, receipt/fiscal, inventory and product combination. It cannot imply support for untested combinations or certify legal, accounting, PCI, security or accessibility status without the relevant qualified assessment.

8. Gate governance and approvers

Gate	Decision	Required approvers	May not self-certify
G0	Verifiable baseline	Program, release, DBA, QA	Implementing squad
G1	Architecture/control freeze	Product, architecture, accounting, inventory, payments, security, country-pack, operations	Implementing engineer
G2	Transaction/access trust	POS/backend, DBA, security, accounting/inventory reviewers	Result-registry author
G3A	Payment truth	Payments, treasury, accounting, security	Provider adapter author
G3B	Document truth	Compliance engineering, country-pack expert, accounting, security	Fiscal-source author
G4	Immediate POS release candidate	Product, store operations, accessibility, security, SRE	POS squad
G5	Store financial close	Store operations, treasury, accounting controller, country-pack	Cashier/declarer
G6	Fulfillment truth	Inventory controller, product, accounting	Fulfillment squad
G7	Financial O2C	Accounting controller, treasury, inventory, product	O2C squad
G8	Operational candidate	SRE, security, accessibility, support and controllers	Release engineer
G9	Production expansion	Executive product owner and all material domain approvers	Any one function

9. Migration, backfill, and rollback strategy

9.1 Standard migration sequence

measure → expand → shadow-write → backfill → reconcile → verify → pilot-read → enforce → contract

- New tables, columns and indexes are additive.
- Backfills are resumable, idempotent, organization-scoped, chunked, observable and hash-certified.
- New code tolerates old and new rows during rollout.
- Uniqueness and not-null constraints follow duplicate/null measurement and evidence-backed resolution.
- Rollback disables new writers/read paths or returns a cohort to observe mode; it does not delete new evidence.
- Immutable financial, stock, receipt, fiscal, reconciliation and audit history is never rewritten.

9.2 Legacy truth handling

Legacy condition	Treatment
Completed sale without result registry	Create historical result-reference from preserved immutable IDs/hashes; do not claim original retry proof
Electronic payment with only operator reference	Mark legacy/unverified or reconciled-by-evidence; never backfill provider capture without provider proof
Sale without immutable fiscal document/source	Materialize only when preserved source provenance is sufficient; otherwise retain explicit evidence gap
Existing journals and stock events	Add source links; never rewrite/delete

Legacy condition	Treatment
Sales-order lifecycle label without transition evidence	Do not synthesize fulfillment history
Existing close/reconciliation report	Preserve and version; do not retroactively certify unsupported sign-off
Fractional historical XAF	Detect, quantify and quarantine for controller-approved remediation; do not silently round history

10. Verification and evidence strategy

10.1 Test pyramid by consequence

Layer	Required coverage
Contract/unit	State transitions, currency/tax/quantity rules, permissions, provider mappings, hashes, redaction, UI state contracts
Service/database integration	Real PostgreSQL transactions, unique constraints, CAS, rollback, source links and balanced postings
Concurrency/chaos	Duplicate click, response loss, simultaneous terminal, webhook/poll race, queue retry, lease expiry and stale aggregate
Cross-domain scenarios	Sale/payment/stock/journal/receipt; order/reservation/fulfillment/invoice/AR; statement/reconciliation/close
Browser/accessibility	Cashier, manager, accountant, support, keyboard, focus, screen reader, touch, responsive, EN/FR and recovery
Provider/hardware	Sandbox/pilot, scanner, printer, drawer, network loss, callback delay and offline replay
Operations	Load/soak, backup/restore, failover, dead-letter, suspense, incident, rollback and cohort expansion

10.2 Mandatory negative scenarios

- Cross-tenant and unauthorized cross-location/terminal identifiers.
- Missing/expired permission, entitlement, assignment, session or fresh authentication.
- Self-approval and expired manager approval.
- Same command/client ID with a different payload.
- Provider pending, unknown, timeout, late capture, duplicate, replayed or forged callback.
- Provider amount, currency or account mismatch.
- Sale transaction failure after provider capture.
- Fiscal worker/authority unavailable, rejected or replayed.
- Receipt delivery unavailable after financial commit.
- Offline revoked device, invalid signature, sequence gap, expired policy or payload conflict.
- Concurrent cart, reservation, goods issue and negative availability.
- Partial delivery, invoice, payment, return and duplicate correction.
- Close with open session/drawer, offline conflict, provider suspense, fiscal dead letter or unposted journal.
- Recovery after browser refresh, process crash, network loss, worker restart, database retry and failover.

10.3 Performance budgets

Journey	Release budget
Local input/touch feedback	p95 ≤ 100ms
Exact scan to acknowledged cart command	p95 ≤ 300ms online; p99 ≤ 750ms
Text search first useful result	p95 ≤ 500ms after 150–250ms debounce
Cart mutation acknowledgement	p95 ≤ 500ms
Tender preflight	p95 ≤ 750ms excluding provider
Financial commit	p95 ≤ 2s excluding provider authorization

Journey	Release budget
Receipt outcome	p95 ≤3s; otherwise explicit retry-required without recharging
Initial interactive POS shell	p75 ≤2.5s on reference device/network
Refresh recovery	Authoritative shift/cart state ≤2s after shell readiness
Catalog DOM	Bounded/virtualized, never proportional to the full catalog

10.4 Standard implementation verification ladder

Run the narrowest focused tests first. At each release-shaped slice, run the applicable equivalent of:

```
npm run typecheck
npm run prisma:validate
npm run service:boundary:fail
npm run policy:gates
node scripts/kontava-moat-release-gate.js --mode fail
npm run build:app
```

Route-facing slices also run the authenticated EN/FR browser matrix. Database migrations and concurrency slices run the real PostgreSQL harness. Provider/device/offline slices run their external/sandbox/reference-hardware suites.

11. POS audit finding traceability

Finding	Implementation ownership	Closure evidence
POS-01 Tender contract contradiction	M1-04; M3A-04 through M3A-07; M4-B03	Method-contract tests, provider sandbox and browser tender matrix
POS-02 Fractional XAF	M1-03; M4-D05; legacy check in 9.2	Schema/service/database/UI property tests and controller review
POS-03 Disconnected offline claim	M1-08; M4-C01 through M4-C03; M8/M9 field proof	Truthful containment, replay/conflict tests and device field evidence
POS-04 Synthetic hardware readiness	M1-08; M4-C04/M4-C05; M9	Adapter tests, freshness states and reference-store hardware evidence
POS-05 Over-privileged cashier seed	M2-B03	Allowlist snapshot and denied-action tests
POS-06 Entitlement inconsistency	M1-06; M2-B01/M2-B02; M9 enforcement pilot	Permission × entitlement × location matrix and browser denied states
POS-07 Draft/cart concurrency	M2-A05 through M2-A07	Real PostgreSQL duplicate/replay/stale-version races
POS-08 Inconsistent stock availability	M1-05; M2-A08; M4-A02/M4-A03	Reserved-stock and concurrent-sale integration/browser tests
POS-09 Incomplete exception workflows	M4-B04/M4-B05; M5; M7	Role-separated E2E for hold, override, refund, void, handover and correction
POS-10 Tablet/mobile cart priority	M4-B01/M4-B02	EN/FR viewport, touch, orientation and virtual-keyboard evidence
POS-11 Accessibility/touch gaps	M4-D01 through M4-D04; M8-07	Automated checks plus keyboard, NVDA, zoom, touch and contrast review
POS-12 Missing inactivity lock/handover	M4-B05	Timer, multi-user, offline, stale-membership and attribution tests
POS-13 Excess customer/receipt exposure	M2-B06; M4-B06/M4-B07	DTO, role, masking, log and screenshot redaction tests
POS-14 Missing performance budget	M4-A02/M4-A05; M4-D06; M8-04	Browser traces, query counts, load/soak and database plans
POS-15 Oversized component/service	M8-06	Characterization tests, graph delta and unchanged domain ownership

Finding	Implementation ownership	Closure evidence
POS-16 Local/offline translation drift	M4-D04	Literal scan, key parity, pseudo-localization and FR review
POS-17 Dense visual hierarchy	M4-B01/M4-B02; M4-D03	Cashier task-time, bright-environment and visual-regression review
POS-18 Incomplete observability/certification	M0; M4-D07; M8; M9	Correlated traces, SLOs, failure injection and immutable release pack

All 18 findings are owned. None may be closed merely because a component was restyled or a unit test was added.

12. Sales-to-Cash blocker traceability

Blocker	Owning milestone(s)
1. No client-result replay	M2
2. No PostgreSQL concurrency proof	M0, M2, M8, M9
3. Manual electronic reference becomes paid	M3A
4. Incomplete canonical provider lifecycle	M1, M3A
5. Provider/statement/settlement/ledger proof not tied to POS	M3A, M5, M9
6. No immutable receipt/fiscal source in every sale	M3B
7. Fiscal helper ownership ambiguity	M3B
8. No production fiscal worker/authority proof	M3B, M9
9. Production receipt-token signing secret absent	M2, M3B, M9
10. Incomplete durable receipt channels	M3B, M4
11. No qualified country-pack approval	M1, M3B, M5, M9
12. Delivery/on-account workflow missing	M6, M7
13. Partial fulfillment/invoice/refund contracts missing	M6, M7
14. UOM/lot/serial/expiry support unproven	M1, M6; unsupported until proven
15. Business day/X/Z/statement aggregate missing	M5
16. Branch reconciliation/sign-off unsupported	M5
17. Inconsistent permission/entitlement contracts	M1, M2
18. Broad module enforcement candidates/missing classifications	M2 triage, M8 observe, M9 narrow enforce
19. Least privilege, inactivity, handover and maker-checker incomplete	M2, M4, M5, M7
20. Store Credit contradiction	M3A, M4
21. XAF minor-unit policy unproven	M1, M4, M9
22. Offline checkout not connected to UI	M4, M8, M9
23. Synthetic hardware readiness	M4, M8, M9
24. Mobile/accessibility/privacy/recovery incomplete	M2, M4, M8, M9
25. No production isolation/load/restore/observability proof	M8, M9
26. No formal certification	External qualified assessments after G8; never self-certified
27. Dirty worktree not tied to a release commit	M0, M9

All 27 blockers remain visible until closed, deliberately unsupported with fail-closed behavior, or assigned to an externally qualified assessment.

13. Historical gap-plan alignment

The historical gap-plan suite remains useful as a platform dependency sequence. Its implementation status must be revalidated against current source before any skill runs.

Historical gap-plan item	Master-roadmap integration
Release verification foundation	M0; hard prerequisite for every later gate
Reconciliation proof launcher	M3A-11
Close invalidation completion	M5-11/M5-12
Workflow Assurance observe pilot	M8-10
Public receipt/API access hardening	M2-B07/M2-B08; verify current signed-token implementation rather than rebuilding it
Provider health/reconciliation operations	M3A-12/M3A-13 and M8-13
Dashboard daily habit/read-model minimization	M8-11/M8-12
Narrow module/workflow enforcement	M9 enforcement progression

If those installed skills are used, retain their prescribed order: gap-plan orchestrator → release verification → reconciliation proof → close invalidation → workflow assurance observe pilot → access boundary hardener → provider health → dashboard daily habit → narrow enforcement. POS-specific access normalization in M2 is a separate current critical-path requirement and must not wait for the broader historical API/receipt hardening skill.

14. Execution packet sequence

Each packet is intended to be independently implementable, reviewable, and evidence-producing.

Order	Packet	Milestone	Required output
1	Verification harness and baseline	M0	Auth browser, PostgreSQL, migration, build and evidence harness report
2	State/command/money/capability freeze	M1	Approved contracts and ADRs
3	POS result registry	M2	Additive schema, finalizer integration and concurrency proof
4	Cart aggregate version/idempotency	M2	Draft/line invariants and stale-conflict proof
5	POS access and least privilege	M2	Unified access descriptor and negative matrix
6	Provider canonical lifecycle	M3A	Versioned adapter and provider-ingestion proof
7	POS provider capture bridge	M3A	Method-aware checkout, clearing/suspense and recovery proof
8	Immutable receipt/fiscal source	M3B	In-transaction source plus worker/delivery evidence
9	POS authoritative read model and preflight	M4	POSSurfaceContext, sellability and result contracts
10	POS workstation, exception, lock and handover	M4	EN/FR responsive/accessibility workflow evidence
11	Offline and device integration	M4	Truthful containment or certified queue/adapters
12	Business day, declaration, X/Z and statement	M5	Close aggregate and provider/accounting handoff proof
13	Delivery order/reservation	M6	Non-posting promise and availability proof
14	Fulfillment/goods issue	M6	Partial fulfillment and inventory/COGS proof
15	Invoice/AR/returns/corrections	M7	Financial O2C and compensating-record proof
16	Operations, assurance and decomposition	M8	SLOs, runbooks, chaos/restore, observe pilot and graph delta
17	Country/provider/hardware pilot	M9	Named production-evidence pack and gate decision

No packet may silently absorb a later packet's scope. A blocked packet leaves a blocker report and does not self-approve a workaround.

15. First 90-day delivery plan

Assuming two-week sprints and two squads:

Sprint	Squad A	Squad B	Shared/control outcome
1	PostgreSQL and result-race harness	Authenticated browser/accessibility harness	G0 registers, pilot scope and baseline evidence
2	Result/state contract design	POS surface/shift/order contract design	G1 contract freeze
3	Result-registry additive schema/shadow write	POS access descriptor and cashier role repair	Migration and authorization review
4	Finalizer result replay and response-loss tests	Cart identity/version/idempotent commands	G2 transaction review part 1
5	Provider canonical state adapter	Inventory sellability/preflight contract	Provider sandbox and stock policy ready
6	Webhook/poll idempotency and payment intent	POSSurfaceContext and privacy-minimized DTO	Security/provider review
7	POS capture bridge and clearing/suspense	Responsive workstation skeleton behind existing commands	Method and UI contract E2E
8	Immutable receipt/fiscal source integration	Shift lock/handover and exception-state UI	Country-pack/receipt review
9	Provider reconciliation/proof and failure cases	Accessibility/localization/device/offline containment	G3A/G3B readiness assessment

At day 90, the target is not production release. The required outcome is G0–G2 passed, G3A/G3B substantially implemented in sandbox, and the new POS shell consuming frozen contracts without weakening the existing cash-sale kernel.

16. Program risk register

Risk	Early warning	Mitigation	Owner
Provider semantics differ from assumptions	State regressions/manual exceptions	Versioned adapter, redacted raw evidence, quarantine and sandbox contract tests	Payments lead
Country-pack decision arrives late	Jurisdiction logic leaks into shared code	Select pilot country in M0; fail closed; block fiscal claims	Product/compliance
Dirty changes collide with core work	Unattributed diff/unstable baseline	Named isolated baseline, file ownership and frequent attribution	Program lead
POS service remains a hotspot	Merge conflicts/regressions	Freeze contracts first; bounded changes; decompose only in M8	Architecture
Backfill invents history	Synthetic captured/fiscal/fulfillment states	Confidence classifications and explicit evidence gaps	Data/controller
O2C duplicates POS kernels	New posting/stock/fiscal finalizer appears	Architecture dependency gate and controller review	Architecture
Module enforcement locks valid tenants	Would-block volume/support spike	Observe, clean false positives, narrow cohort and tested rollback	Module platform
Offline breadth outruns control	Duplicate/provisional/fiscal conflicts	Narrow supported commands and field proof before claims	Offline lead
Device readiness remains cosmetic	Unknown device failures appear as ready	Adapter freshness contract and hardware pilot	Device lead
Maker-checker harms small-store flow	Abandonment/credential sharing	Materiality thresholds and measured pilot	Product/security
Evidence leaks sensitive data	Raw callback/customer/payment data in logs	Redaction schemas, fixtures, DLP/secret scan	Security

Risk	Early warning	Mitigation	Owner
Static gates mask runtime failures	CI green but staging incidents	Evidence taxonomy and required real database/provider/browser gates	Release/SRE
Scope expands to generic ERP	Gate durations and backlog grow	Non-goals, change control and outcome-owned milestones	Product
Accessibility deferred to polish	Late layout/workflow rework	Accessibility contract and tests begin in M0/M1, not M8	Accessibility lead

17. Operating metrics

Transaction integrity

- Sale attempts, registry claims, original-result replays, payload conflicts and abandoned claims.
- Duplicate-consequence detector: target zero.
- Commit latency and rollback/error rate by tender, location and terminal.

Payments and treasury

- Pending/unknown age, late capture and captured-without-sale cases.
- Callback latency, settlement match rate, fee variance, suspense value/age, dispute and chargeback counts.
- All amounts segmented by organization, provider account, currency and business date.

Receipt/fiscal

- Completed-sale immutable-source coverage: target 100%.
- Fiscal queue age, rejection, dead letters and sequence gaps/duplicates.
- Delivery success/retry/failure by channel without destination PII.

Inventory and O2C

- Stale-stock conflicts, reservation aging, oversell attempts, partial fulfillment and backorder age.
- Goods-issue-to-invoice lag, stock/COGS tie-out and return-disposition age.
- Order-to-cash cycle and AR aging.

Store close

- Sessions closed on time, variance frequency/value and pending approvals.
- Business-day blocker age, statement-posting lag and reconciliation completion.
- Late correction and close-invalidation frequency.

Experience and reliability

- Scan/search/cart/preflight/commit latency and recovery completion.
- Offline conflict and device-failure rate.
- Critical accessibility regression count and cashier task success.
- SLO/error-budget, worker backlog, restore/failover and support-case resolution.

Metrics explain and route work; they never become a second financial or stock truth.

18. Program stop/go rules

Stop the affected cohort or phase when any of the following occurs:

- Duplicate financial, stock, receipt/fiscal, statement or result consequence.

- False successful payment from pending/unknown/timeout/manual evidence.
- Cross-tenant or unauthorized cross-location/terminal access.
- Unbalanced or unexplained posting/stock/provider/statement difference.
- Lost immutable source, audit or reconciliation evidence.
- Unrecoverable offline/device/provider/fiscal/close conflict.
- Sensitive payment data, secrets or unnecessary customer PII in logs/evidence.
- Critical accessibility barrier in a supported cashier/approval workflow.
- Migration/backfill cannot reconcile or rollback safely.
- Error-budget, support-capacity or suspense-aging threshold breach.

Restart requires root-cause evidence, corrective change, focused regression, full affected gate rerun and explicit approver decision.

19. Definition of complete

The master roadmap is complete only when:

1. G0–G9 pass for an explicitly bounded supported configuration.
2. All 18 POS findings and 27 Sales-to-Cash blockers are closed, fail-closed unsupported, or assigned to qualified external assessment.
3. The historical gap-plan dependencies used by this program are either revalidated as complete or closed with current evidence.
4. Immediate POS and delivery/on-account orchestration reuse kernels without duplicate finalizers or posting/stock/fiscal owners.
5. Every terminal financial, inventory, payment, receipt/fiscal, statement, fulfillment and correction fact is immutable, source-linked, scoped, authorized, idempotent, auditable and recoverable.
6. The release evidence pack is bound to the deployed commit, migrations, configuration and environment.
7. Operators can see unsupported, degraded, stale, blocked and recovery states without false readiness.
8. Remaining limitations are visible in product packaging, support documentation and customer communication.

Until those conditions are met, the correct program status is:

Implementation in progress — not production certified.